Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates reactions, maximum moment and shear, and maximum deflection of a single-span beam for the standard support and load cases. Left end at x = 0; loads downward; sagging moments and downward deflections positive.

WHAT IT CALCULATES	RELATION
Formula source	Roark Table 8.1, in this sheet's own notation, for the chosen support and load case
Simply supported beam, point load at a (b = L − a), reaction and largest moment	$R_A = F\,b / L$; $M_{max} = F\,a\,b / L$
Simply supported beam, point load at a, largest deflection when a ≥ b	$\delta_{max} = F\,b\,(L^2 - b^2)^{(3/2)} / (9\sqrt{3}\,L\,EI)$ at $x = \sqrt{((L^2 - b^2)/3)}$
Propped cantilever, uniform load, reaction and moments	$R_B = 3wL/8$; $M_A = -wL^2/8$; sagging $M_{max} = 9wL^2/128$ at $x = 5L/8$

Basis: Shear, moment, slope and deflection formulas for elastic straight beams[1](Table 8.1, 8th ed.).

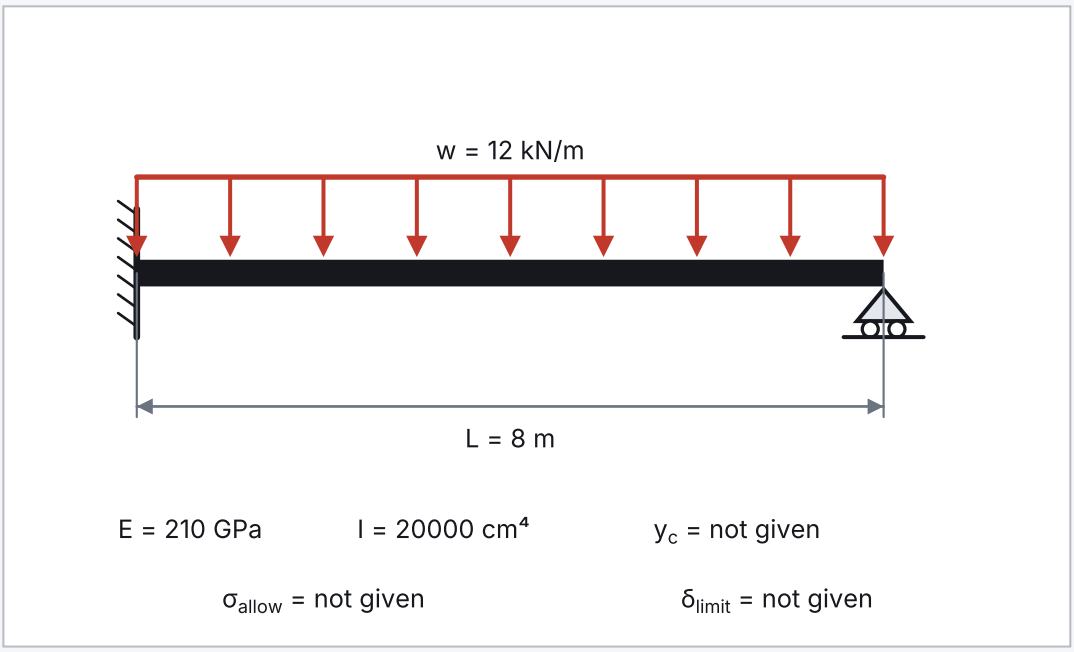


Figure 1 — Propped cantilever, fixed left and pinned right, uniformly distributed load w. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
case	Case	propped cantilever (fixed left, pinned right), UDL		assumption: Fixed at the wall, propped at the free end.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Span	8	m	drawing: DRG-CA-033 rev A	0.1 … 100 m
I	Second moment of area	20000	cm^4	reference: Fabricated section, S1 sheet	1 … 100000000 cm^4
E	Young's modulus	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
w	Distributed load, downward (optional; blank = not used by the point-load and couple cases)	12	kN/m	spec: Load schedule LS-03	0 … 10000 kN/m

3. Assumptions

A1. The beam is prismatic, linear elastic and Euler–Bernoulli, with small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as propped cantilever (fixed left, pinned right), UDL. (Fixed at the wall, propped at the free end.)

4. Working

[5] EI = E * I
= 210 GPa × 20000 cm^4
= 42000 $kN·m^2$
Flexural rigidity

[7] R_A = 5 * w * L / 8 [1](8.1 2c)
= 5 × 12 kN/m × 8 m / 8
= 60 kN
Fixed-end reaction

[8] R_B = 3 * w * L / 8
= 3 × 12 kN/m × 8 m / 8
= 36 kN
Propped-end reaction

[9] M_{min} = -w * L^2 / 8
= -12 kN/m × (8 m)^2 / 8
= -96 kN·m
Fixed-end (hogging) moment

[10] M_{max} = 9 * w * L^2 / 128
= 9 × 12 kN/m × (8 m)^2 / 128
= 54 kN·m
Maximum sagging moment

[11] x_M = 5 * L / 8
= 5 × 8 m / 8
= 5 m
Position of the maximum sagging moment

[12] V_{max} = 5 * w * L / 8
= 5 × 12 kN/m × 8 m / 8
= 60 kN
Maximum shear

[13] x_{delta} = L * (15 - sqrt(33)) / 16
= 8 m × (15 - sqrt(33)) / 16
= 4.6277 m
Position of the maximum deflection, from the fixed end
[14] δ_{max} = w * L^4 / (185 * EI)
= 12 kN/m × (8 m)^4 / (185 × 42000 kN·m²)
= 6.3259 mm
Maximum deflection

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	60 kN	Left reaction
R_B	36 kN	Right reaction
M_{max}	54 kN·m	Maximum sagging moment
M_{min}	-96 kN·m	Minimum (hogging) moment
V_{max}	60 kN	Maximum shear
δ_{max}	6.326 mm	Maximum deflection
x_M	5 m	Position of M_{max}
x_{delta}	4.628 m	Position of δ_{max}

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

Template struct.beam-formulas@1.2.0 Content hash f598af29a4da6e15 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.